实验代码与求解器比较

工作顺序与项目框架 · C01 · 保留原主线

两套系统、七类方法、四组实验场景

Retain C01-C08. C09-C14 add comparisons of Jacobi, GS, SOR, CG, GMRES, AMG and GMG, with a direct baseline and separate multigrid preconditioning tests. X01-X04 remain model extensions. All tables and figures are planned outputs; no experiments are claimed to have been run.

阶段	任务安排	理论依据
原主线	C01-C08: data, assembly, verification, convergence and baseline comparisons	T01-T08
新增求解器	C09-C11: operator platforms, Krylov methods and multigrid	T14-T19
新增实验	C12-C14: grid scale, geometric hierarchies, multiple right-hand sides and applicability	T20
模型拓展	X01-X04: FEM, L-shaped domain, local refinement and pure Neumann data	T09-T13

[ ] C01 扩充模块、配置与统一接口

如何完成：Retain the problem, grid, assembly and metrics modules. Add CG and GMRES to solvers, and AMG/GMG setup, V_cycle and as_preconditioner to multigrid. Record the operator platform, restart length, smoother, coarse solver, resource limits, mesh, boundary closure and tolerance in one configuration, together with software versions, hardware and thread counts.

完成标志：One configuration entry point; each case_id identifies the operator, solver, preconditioner and parameters.

建议模块	职责
operator_platforms	Retain original A,b; construct equivalent B,c; verify matrix conditions.
solvers / multigrid	prepare → solve → statistics; reuse setup where permitted.
metrics / experiments	Validate against the original equation; run and export S1-S4.

CG requires SPD. The initial multigrid comparison also uses SPD configurations; retain the original nonsymmetric platform separately. See C09-C14 for the solver chapter. The referenced iterative_solvers_skeleton.py was not supplied; verify its interface before integration.

数据与四种系统装配

内部离散共享，只切换边界模块 · C02-C03

[ ] C02 实现解析数据、网格与边界标签

如何完成：Implement $u_*=\exp(x+y)$, $f=-2\exp(x+y)$, $g=u_*$ on Dirichlet nodes, $q=\exp(1+y)$ and $r=3\exp(1+y)$. Check $q>0$ and $r=q+2u_*$. Label the three Dirichlet edges and all corners before the open right edge. Use $h=1/N$ and unknowns $i=1,\ldots,N$, $j=1,\ldots,N-1$, with $idx=(j-1)N+i-1$; always keep coordinate and label arrays consistent with that ordering.

完成标志：Data functions and small-grid label/index plots; verify $m=N(N-1)$ unknowns.

[ ] C03 装配 Neumann / Robin × 一阶 / 虚点系统

如何完成：Use the same interior five-point stencil in all four cases: diagonal $4/h^2$ and four cardinal neighbors $-1/h^2$. Move a known neighbor to the right-hand side using $b_k \leftarrow b_k - a_{k\ell} g_\ell$. At each open-right-edge node, insert exactly one boundary row from the table. Eliminate every ghost value analytically; it never enters the unknown vector.

完成标志：A common assemble(config) → A,b,meta interface; only the right-edge module changes between cases.

右边界行	对角与向内系数	纵向系数与初始右端
Neumann + 一阶	$1/h$; $-1/h$	None; q_j
Robin + 一阶	$1/h+\beta$; $-1/h$	None; r_j
Neumann + 虚点	$4/h^2$; $-2/h^2$	Each $-1/h^2$; $f_{N,j}+2q_j/h$
Robin + 虚点	$4/h^2+2\beta/h$; $-2/h^2$	Each $-1/h^2$; $f_{N,j}+2r_j/h$

$\beta=2$. The inward neighbor is $(N-1,j)$; vertical neighbors are $(N,j\pm1)$. A known corner in a ghost row adds g/h^2 to the RHS. Table entries precede these known-value contributions.

装配原则

Right-edge rows cover only $j=1,\ldots,N-1$. The three Dirichlet edges and all corners remain prescribed. Interior rows at $i=N-1$ still connect to right-edge unknowns. See T03-T05.

小网格核验与直接基准

先核验装配，再运行网格序列 · C04-C05

[ ] C04 检查系数、符号与精确采样缺陷

如何完成：Start with N=4. Print representative interior rows, rows next to Dirichlet data, right-edge rows and rows next to corners, then compare every coefficient with T03-T05. Compute $\tau = A u_* - b$ using exact nodal samples. On nested grids, inspect defects at the fixed location y=1/2 and check their predicted size before launching larger cases.

完成标志：Coefficient-check records and a substitution table; unresolved assembly errors block batch experiments.

未缩放行	待核验的局部缺陷
内部 PDE 行	$O(h^2)$; leading term $-h^2 u_*/6$.
一阶右边界行	$O(h)$; leading term $-hq/2$.
虚点右边界 PDE 行	$O(h)$; leading term $+hq/3$. Multiplying the whole row by h^2 gives $O(h^3)$.

τ is a discretization defect and need not vanish. The computed-solution residual is r=b−AU. Store and interpret these two quantities separately.

[ ] C05 保存直接参考解与矩阵诊断

如何完成：For each boundary/closure combination and mesh, perform one sparse direct baseline solve and save U_{dir}. Evaluate the common scaled residual ρ_s defined in T07. Plot the sparsity pattern of the original A and record m, nnz and $\delta_A = \|A - A^T\|_F / \|A\|_F$. Retain these data as the reference for subsequent error and solver comparisons.

完成标志：Reference solutions, direct-solve times, scaled residuals, four sparsity plots and an asymmetry table.

Direct timing includes factorization and solution; time assembly and plotting separately. Record the actual ordering and backend rather than equating library defaults with theoretical nested dissection. If LU fill is studied, obtain factor statistics explicitly and explain any timing changes. See T05 and T07.

收敛实验与分区误差

四种组合、至少四网格 · C06 · 主线实验

[ ] C06 运行网格收敛实验并分析误差来源

如何完成：Begin with N=16,32,64,128 and run Neumann/Robin with first-order/ghost closures, giving 16 baseline systems. For every system, compute L^2 and maximum errors globally, in the interior and on the right edge. Examine neighboring-mesh rates before selecting a fit interval. Use regional errors to explain whether the boundary treatment limits global accuracy.

完成标志：One convergence table per combination, global/regional curves and an interpretation of boundary influence.

区域	L^2 权重与节点集合	另报指标
全局	$\sqrt{(h^2\Sigma e^2)}$; $1\le i\le N$, $1\le j\le N-1$	Maximum nodal error
内部	$\sqrt{(h^2\Sigma e^2)}$; $1\le i,j\le N-1$	Maximum nodal error
右边界	$\sqrt{(h\Sigma e^2)}$; $i=N$, $1\le j\le N-1$	Maximum nodal error

识别渐近收敛区间

Use $p=\log(E_h/E_{h/2})/\log 2$. If the finest rates still drift, refine further or state that the asymptotic regime has not been reached. Do not select only mesh pairs that match a prediction.

计划产物	字段或图示内容
全局误差表	N, h, m, $E_{2,h}$, E_∞, p_2, p_∞, ρ_s
分区误差表	$E_{2,I}$, $E_{\infty,I}$, $E_{2,B}$, $E_{\infty,B}$ and their local rates
收敛图	Error versus h on log-log axes; identify the norm and closure.
诊断图	A representative error heatmap and right-edge/interior comparison.

Hypotheses to test: first-order closure gives about order one in right-edge L^∞; ghost closure gives about order two in both global norms. These are predictions, not numerical acceptance targets. See T06.

定常迭代与公平计时

统一停止条件，核验实际精度 · C07-C08

[ ] C07 实现 Jacobi、GS 与 SOR

如何完成：Use the corrected splitting formulas in T08 and start all unknown vectors from zero. Jacobi reads the previous vector; GS and SOR use components already updated during the current sweep. Recompute the true scaled residual after each complete sweep. Return the residual history, iteration count and status, with a fixed traversal order, maxiter and divergence checks.

完成标志：Three interchangeable solvers with documented sweep order, stopping logic and failure states.

For A=D−E−F, the E and F terms on the Jacobi/GS right-hand sides have positive signs. Restrict SOR candidates to $0<\omega<2$, but verify convergence rather than assuming it. Record tuning costs separately.

[ ] C08 匹配精度比较并导出复现材料

如何完成：Use the same ρ_s for every solver, starting with tol=10^{-10}. Check the direct-reference solution-difference budget separately for every combination and mesh. Tighten tolerances when needed and adopt a common final comparison target. Record assembly, setup and solve times separately, together with iteration counts; include residual-checking costs in solve time and retain failures.

完成标志：Comparison tables, time/iteration plots, CSV files and one reproduction command; state the tested range if no crossover is observed.

$$\|U_{it}-U_{dir}\|_\infty \le 0.01\, E_{\infty,dir}$$

$E_{\infty,dir}$ is the direct solution's nodal discretization error against u_*. A fixed residual tolerance alone does not guarantee this 1% budget.

记录类别	必备内容
每次运行	case_id, solver, ω, tol, iters, ρ_s, solution difference, setup_s, solve_s, status
公平性说明	Initial guess, traversal order, implementation, hardware and threads; identify Python-loop overhead.
复现材料包	Configuration, versions, raw CSV, plotting scripts, README and run entry point.

Do not preselect a timing ranking or crossover. Model iteration counts have positive slopes versus N; measure the mixed-boundary behavior. Keep failed-run statuses instead of replacing them with zero errors or blank cells. See T08.

两个独立的模型拓展

矩形 FDM-FEM 基准与 L 形域 · X01-X02

[ ] X01 搭建矩形 FDM / FEM 公平对照〔新增拓展〕

如何完成：Create a separate smooth square problem with Dirichlet data on every edge and a shared exact solution and nodal grid. For FEM, split each square along a diagonal into two right triangles and assemble P1 elements. Match nodal norms and solution accuracy; record assembly and solve times separately while preserving each method's load construction.

完成标志：Degrees of freedom, actual nnz, nodal errors and separate timings; optionally add FEM's continuous integral L^2 error.

Verify $A_{FEM} \approx h^2 A_{FDM}$ on interior rows. Remove explicitly stored zeros before counting nnz. Do not assume 7- or 9-point couplings; explain differences in the load vectors. See T09.

[ ] X02 搭建 L 形光滑与奇异基准〔开放方向〕

如何完成：On meshes fitting the L-shaped domain, test the smooth solution $e^x \cos y$ and the singular solution $r^{2/3}\sin(2\varphi/3)$, both with f=0 and analytical Dirichlet data. For the singular solution, implement the atan2 branch correction and verify both zero-trace rays. Set only the singular solution to zero at the origin; never evaluate its gradient there.

完成标志：Mesh and boundary checks, two error heatmaps and convergence tables explicitly labeled FDM or FEM.

明确选择并标注离散路线

FDM: use an aligned five-point grid; report discrete L^2, difference-gradient energy and nodal maximum errors with explicit edge sets and weights. Label rates as measured FDM results. FEM: use P1 triangles and full-domain continuous L^2/energy errors to establish the uniform baseline for local refinement. These datasets are not interchangeable.

Use a base mesh plus at least four uniform refinements: $\ell=0,\ldots,4$ gives five levels. For FDM on the L-shaped domain, h=2/N and N=16,…,256 is one choice; record FEM triangle-refinement levels separately. See T10.

Implement the exact gradient for FEM error integration. Retain corner elements, use suitable subdivision/quadrature for the singularity and check integration accuracy. Save h, degrees of freedom, nnz, errors and time at every level; plot error versus both h and degrees of freedom.

局部加密与纯 Neumann 数据

独立拓展与主线核对 · X03-X04

[ ] X03 实现分级或自适应局部细化〔开放方向〕

如何完成：Reuse the FEM uniform baseline from X02. For graded meshes, refine to the target size law in T11. For adaptivity, apply solve → normal-flux residual indicator → Dörfler marking → NVB or red-green refinement. Fix θ, initially 0.5 if desired, and verify conformity and element shape quality after refinement. Compare against uniform meshes at matched norms and solution accuracy.

完成标志：Uniform/local mesh plots, full-domain L^2 and energy errors versus degrees of freedom, indicators, nnz and timing tables.

On nonuniform meshes, do not replace integration by a single h^2 nodal weight. Use the normal jump $[[\nabla u_h \cdot n_E]]$, not the tangential jump. Include every corner element in the error integrals. See T11-T12.

[ ] X04 验证纯 Neumann 兼容性与规范条件〔选做〕

如何完成：Build a separate problem with prescribed flux on the entire boundary, first defining all four edges and corner rows. Verify $A1 \approx 0$ and $w^T A \approx 0$, and measure the compatibility defect $w^T b$. If the RHS is corrected, retain the correction. After enforcing compatibility, fix the mean or one node and align the exact solution to the same gauge.

完成标志：A full-boundary data table, compatibility defects, the nullspace before gauging and errors after gauging. See T13.

原主线提交前核对	完成
Four boundary rows and exact-sample defects checked; corners correctly assigned.	□
Four combinations on at least four meshes; six errors, fit intervals and scaled residuals saved.	□
Sparsity plots, original asymmetry and direct/iterative matched-accuracy evidence saved.	□
Times, counts, statuses, crossover assessment and reproduction materials retained.	□

Reorganized from the supplied problem screenshots, modeling draft and corrections. The added FEM benchmark and open directions remain separate from the core. All tables and figures are planned outputs; the solver extensions continue in C09-C14.

用于公平比较的两套平台

C09 | 同一离散问题，两种等价表示

[ ] C09 保留 A，并构造等价 SPD 系统 B

如何完成：Use T14: W=1 on interior rows, 1/h on first-order boundary rows, and 1/2 on ghost boundary rows. Form B=WA and c=Wb. Check symmetry, verify small instances by Cholesky or eigenvalues, and retain the SPD proof. Confirm agreement with the original direct solution.

完成标志：An operator interface selecting A or B, while preserving original A,b for all final checks.

This is equation rescaling: multiply both sides of each equation by a positive weight, without changing its solution. It creates the common SPD platform needed for CG. It does not justify applying standard CG to original A.

平台	参加的方法	目的
原始 A	Direct, Jacobi, GS, SOR and GMRES.	Preserve the nonsymmetric problem.
共同 SPD 平台 B	Direct + all seven independent methods + CG+AMG and CG+GMG.	Make the primary comparison on one matrix.
SPD 的 FEM 刚度 K	Direct, CG, GMRES, AMG, GMG and PCG combinations.	Test geometric and local-mesh scenarios; no W needed.

使用十个含义明确的方法标签

Seven independent methods + one direct baseline + two PCG combinations = ten. Label AMG-V and GMG-V as standalone iterations. Label CG+AMG and CG+GMG separately; do not merge both roles into an ambiguous AMG or GMG result.

统一接口： prepare(operator,config) → setup; solve(setup,rhs,x0) → U,history,stats. Save platform, solver, preconditioner and hierarchy_id. Record the W transformation cost separately. For FEM, K,d is the original equation.

实现克雷洛夫子空间求解器

C10 | CG、GMRES 与统一精度核验

[ ] C10 接入 CG 与重启 GMRES

如何完成：Use CG only on B or an SPD FEM K. GMRES may use A, B or K. Start with restart=30, then vary it in a separate sensitivity test. Explicitly record the library version, rtol, atol, callback type and maxiter meaning. Count Arnoldi steps separately from restart cycles.

完成标志：Solver wrappers, residual histories, work counters and unambiguous exit status.

库容差只是内部阈值

After the library returns, recompute the original-equation scaled residual and the direct-solution difference. If either misses the common target, tighten the internal threshold and continue. Include all retries, steps and elapsed time in the formal result.

GMRES may return after one restart cycle so that the current iterate is available for a true-residual check. A preconditioned residual callback may be useful for diagnostics, but it is not the common stopping measure.

实现事项	必要规则
CG / PCG 算子	Validate the SPD formulation and preconditioner assumptions.
GMRES 资源预算	Record inner steps, restart length and completed restart cycles.
初始猜测	Use the same zero initial guess unless a separately declared warm-start test is run.
失败处理	Predeclare iteration, time and memory budgets; save failures and timeouts.

The mathematical assumptions are in T15-T16. A fixed nonsymmetric preconditioner is a GMRES option; a changing or nonlinear action requires a suitable flexible method and is outside the required baseline.

实现两种多重网格方法

C11 | 网格层级、V 循环与两种求解角色

[ ] C11 构建可配 PCG 的 AMG / GMG 循环

如何完成：For square GMG, use levels N=4,8,...,N_{fine} with bilinear prolongation. For nested L-shaped P1 FEM, use finite-element space transfers. Build AMG interpolation by smoothed aggregation. Use fixed Galerkin V(1,1) cycles, paired forward/backward GS, and an SPD direct coarsest solve.

完成标志：Reusable hierarchies, apply_V from zero correction, standalone MG solvers and PCG preconditioner interfaces.

组件	首次比较配置
转移与粗化	GMG: geometric or FEM prolongation. AMG: aggregation-based P. Use $R=P^T$ and $B_c=P^TB_fP$.
平滑	One forward-GS pre-sweep and one backward-GS post-sweep.
最粗层	Target at most 32 unknowns; fixed SPD direct solve.
独立求解	Repeat V-cycles; check the common true residual after each cycle.
PCG 预条件	Use the same fixed, linear, SPD V-cycle for every preconditioner call.
统计记录	Save level sizes, transfer dimensions, hierarchy ID, C_{op} and C_{grid}.

保持比较口径清楚

Reuse the same hierarchy for standalone MG and its PCG combination whenever possible. Do not change smoothing counts, inner tolerances or the hierarchy during standard PCG. If the SPD conditions fail, mark that combination inadmissible and use a separately labeled GMRES configuration.

One V-cycle and one CG step have different costs. Save both outer iterations and preconditioner/V-cycle calls, and compare actual time and matrix-vector work. T17-T19 explain the components and compatibility requirements.

一次受控实验如何进行

比较流程 | S1-S4 的执行逻辑

Each run fixes one matrix, right-hand side, initial guess, accuracy target and resource budget. Only the solver label changes. Repeat this procedure across the four scenarios rather than combining different matrices into one unexplained ranking.

步骤	操作与原因
1. 固定问题	Select the mesh, boundary type, closure and platform. Preserve the original equation.
2. 建立参考	Obtain U_{dir} and the manufactured-solution error E_{disc}.
3. 准备方法	Time factors, preconditioners or hierarchy construction separately.
4. 求解并核验	Use the common scaled residual and check the algebraic-error budget.
5. 记录成本与状态	Save setup, solve, total time, memory, work counters and any failure.

$$E_{disc}=\|U_{dir}-u^*\|,\ E_{alg}=\|U_{method}-U_{dir}\|$$

先匹配精度，再比较成本

Start with the common scaled-residual target 10^{-10}, and verify $E_{alg}\le 0.01E_{disc}$. A fixed library tolerance alone does not guarantee this budget. For mixed-boundary runs use the stated nodal norms; in FEM include continuous L^2 and energy checks.

Even when solving B, evaluate residuals with original A,b. For FEM use original K,d. Fully converged solvers should agree on the discrete solution; time, memory and robustness distinguish how efficiently they reach it.

Each scenario tests a condition: S1 size, S2 nonsymmetry/row scaling, S3 mesh hierarchy, S4 repeated right-hand sides. The following tasks specify the runs.

S1-S2：规模与非对称性

C12 | 比较七类方法，分清影响因素

[ ] C12 运行网格规模与行缩放对照

如何完成：On shared B, run the ten labels for the four boundary/closure combinations and N=16,32,64,128. On original A, retain direct, Jacobi, GS, SOR and GMRES. Add GMRES(A) versus GMRES(B) to separate the benefit of row scaling from a change of algorithm.

完成标志：Same-platform cost tables, A/B bridge plots, size trends and documented failure ranges.

实验	固定与变化因素	待回答问题
S1：规模	Fixed accuracy and formulation; vary N. Extend to 256 and 512 if feasible.	How do work, memory and total time grow?
S2：系统表示	Same physical discrete problem; compare GMRES(A) and GMRES(B).	What does row scaling contribute?
S2：边界	Separate Neumann/Robin and first-order/ghost results.	Does boundary treatment affect convergence?
GMRES 敏感性	On a representative grid use restart=20,30,50,100; keep other choices fixed.	How do basis cost and stagnation trade off?

核心实验的运行规模

4 combinations × 4 meshes × 10 labels = 160 run groups on the common platform. Predeclare common resource limits. Record capped or failed runs honestly. On larger grids, compare the jointly completed range and show censored results as well.

Apply the same original-equation residual and $\|U-U_{dir}\|_\infty \le 0.01 E_{\infty,dir}$ check to every run. Record PDE errors alongside algebraic errors. Different final accuracies must not create an artificial timing advantage.

This experiment investigates whether multigrid setup pays off as size grows. It does not require AMG, GMG or any other method to win.

S3-S4：网格层级与重复求解

C13 | 考察几何层级与准备工作复用的收益

[ ] C13 比较网格场景与多右端求解

如何完成：For S3, use L-shaped P1 FEM on nested uniform levels and then on the same local/adaptive meshes for every participating method. For S4, hold one admissible operator fixed and change only its right-hand side. Reuse each method's factors or hierarchy; start each solve from zero.

完成标志：Mesh-case comparisons, single/batch costs and evidence about when setup reuse is worthwhile.

场景	参加方法与控制条件	待回答问题
S3a：均匀 FEM	Direct, CG, GMRES, AMG, GMG and both PCG combinations; retain levels.	What does a ready geometric hierarchy offer?
S3b：局部 / 自适应	Use the same conforming mesh. GMG participates only with reliable levels and transfers.	Do local meshes degrade convergence or increase maintenance cost?
S4：固定矩阵	R=1,10,50; one setup per method; reuse LU factors or MG levels.	Does amortization change the method choice?

Mark GMG as not configured when transfers are unavailable. For S3, also require the algebraic difference in continuous L^2 and energy norms to be below 1% of the corresponding direct-solution discretization error.

矩阵不变，生成不同右端

For the square problem use $u_s=e^{x+y}+a_s\sin(\pi x)\sin(\pi y)$, with $a_s=0.2s/50$ and s=1,...,50. Recompute manufactured f,g,q,r. With mesh, boundary type and β fixed, only b changes; A,W,B remain fixed. Each batch uses the same first R cases.

$$T_{batch,solver}(R)=T_{setup}+\Sigma_{s=1\ldots R}T_{solve,s};\ average=T_{batch,solver}/R$$

Report assembly and row transformation separately in end-to-end cost. LU must reuse its factorization instead of calling a fresh factor-and-solve for every RHS. Each RHS has its own direct reference and error budget.

测量、解释与复现

C14 | 将结果转为有适用条件的方法选择指南

[ ] C14 导出证据并解释方法适用场景

如何完成：Export S1-S4 with platform, accuracy and resource budgets fixed. Write each conclusion as: conditions, accuracy evidence, setup/solve/memory cost, then recommendation and limitation. Preserve failures and not-configured entries instead of declaring a universal winner.

完成标志：A seven-method selection table, raw CSV files, figures, configurations and a reproducible entry point.

类别	必须记录的内容
精度	Original scaled residual, direct-solution difference, status and failure reason; six mixed-boundary errors or relevant FEM norms.
时间	Assembly, W transformation, setup, solve and single-RHS total; residual checks are included in solve.
工作量	Sweeps; CG steps; GMRES inner steps/restarts; V-cycles; preconditioner calls; matrix-vector products.
内存与参数	Peak memory and measurement scope; restart length; LU fill; levels and hierarchy complexity.
复用	RHS count, factor/hierarchy reuse flags, hierarchy ID, batch total and average cost.

必备图表

Matched-accuracy total time versus degrees of freedom; iterations/cycles versus mesh size; true residual versus time or matrix-vector work; setup/solve breakdown; peak memory; and average batch cost. Different units of iteration must not be merged into an unexplained comparison.

Single-RHS cost includes necessary setup; only repeated solves amortize it. State whether memory is process RSS or an array estimate. Record implementation language and thread settings. Explain timing variability without selecting only the fastest run.

实现参考： SciPy CG; SciPy GMRES; PyAMG aggregation; PyAMG multilevel statistics; PETSc PCMG. Record installed versions when implementing. No numerical ranking is claimed here.